

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 12B

LATTICE ENERGY (BORN-HABER CYCLES, COVALENCY & HYDRATION)

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 12: ENTROPY AND ENERGETICS
SUB-TOPIC:	12B: Lattice Energy (Born-Haber Cycles, Covalency & Hydration)
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	202 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

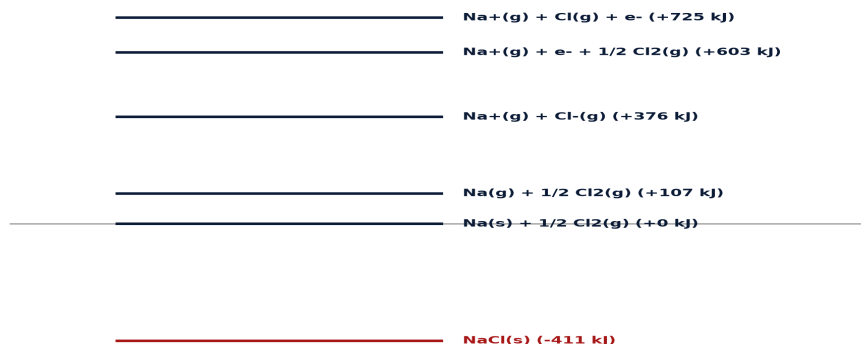
SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25

Q1. 1. Lattice Energy Definition

[3 Marks]
WCH14/01/Jan23/Q7

Lattice energy is a measure of the strength of ionic bonding in a crystal lattice.

Born-Haber Cycle for Sodium Chloride (NaCl)



(a) Define standard lattice energy (lattice formation enthalpy). [2]

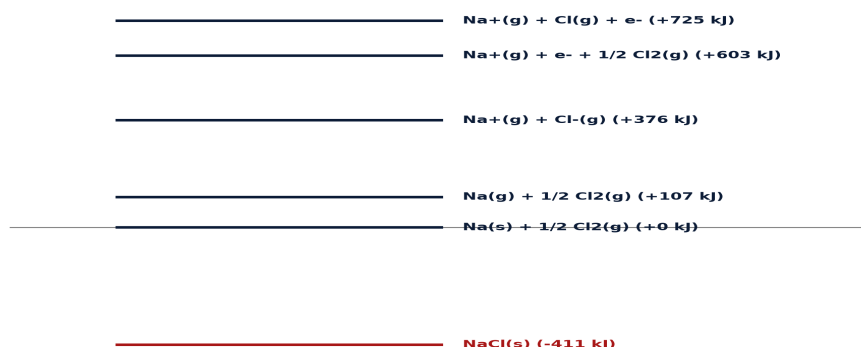
(b) State whether lattice energy is endothermic or exothermic. [1]

Q2. 2. Born-Haber Cycle Calculation for NaCl

[5 Marks]
WCH14/01/Oct22/Q8

Data (kJ mol⁻¹): $\Delta_f H(\text{NaCl}) = -411$, $\Delta_{\text{atom}} H(\text{Na}) = +107$, 1st IE(Na) = +496, $\Delta_{\text{atom}} H(\text{Cl}) = +122$, 1st EA(Cl) = -349.

Born-Haber Cycle for Sodium Chloride (NaCl)



(a) Construct a Born-Haber cycle for sodium chloride. [2]

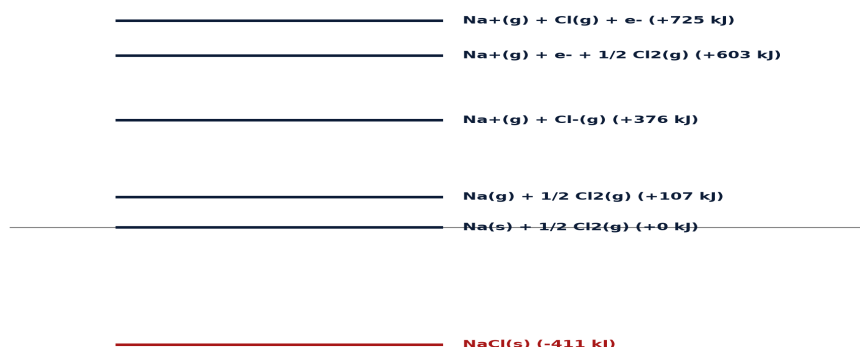
(b) Calculate the lattice energy of sodium chloride, $\Delta_{\text{LE}} H(\text{NaCl})$. [3]

[3 Marks]

Q3. 3. Born-Haber & Lattice Energy Step 3

WCH14/01/Sample/Q3

Consider ionic compound 3 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

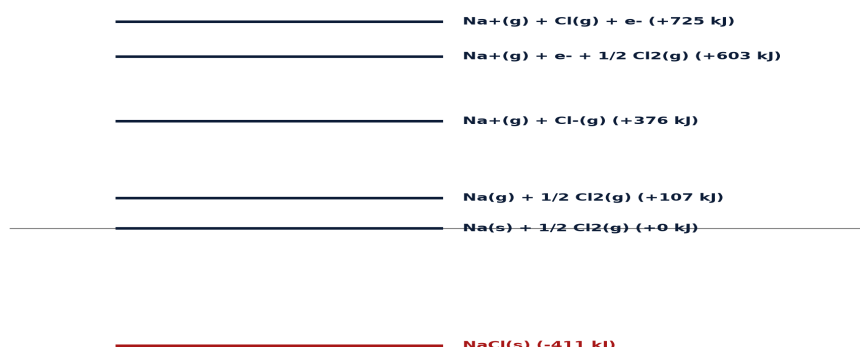
(b) Compare the lattice energy of MX with M_2X (where M is divalent). [1]

[3 Marks]

Q4. 4. Born-Haber & Lattice Energy Step 4

WCH14/01/Sample/Q4

Consider ionic compound 4 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

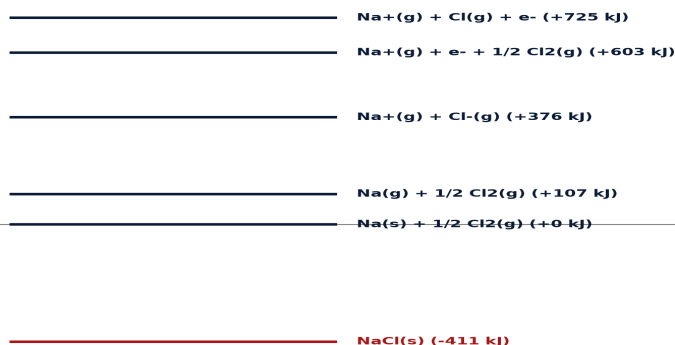
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M_2X (where M is divalent). [1]

Q5. 5. Born-Haber & Lattice Energy Step 5**[3 Marks]****WCH14/01/Sample/Q5**

Consider ionic compound 5 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

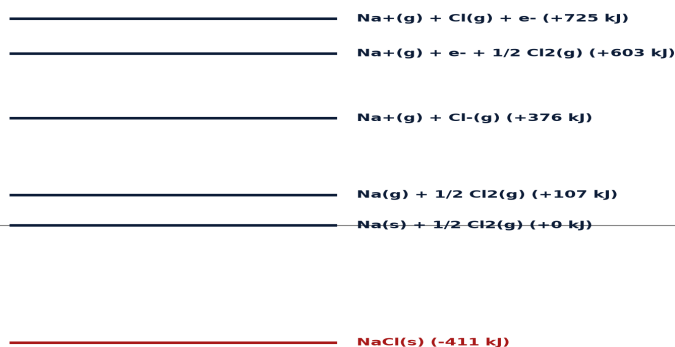
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M_2X (where M is divalent). [1]

Q6. 6. Born-Haber & Lattice Energy Step 6**[3 Marks]****WCH14/01/Sample/Q6**

Consider ionic compound 6 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

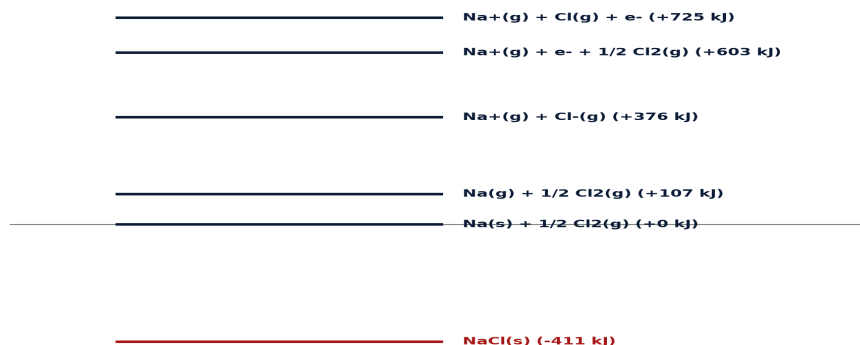
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M_2X (where M is divalent). [1]

Q7. 7. Born-Haber & Lattice Energy Step 7**[3 Marks]****WCH14/01/Sample/Q7**

Consider ionic compound 7 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

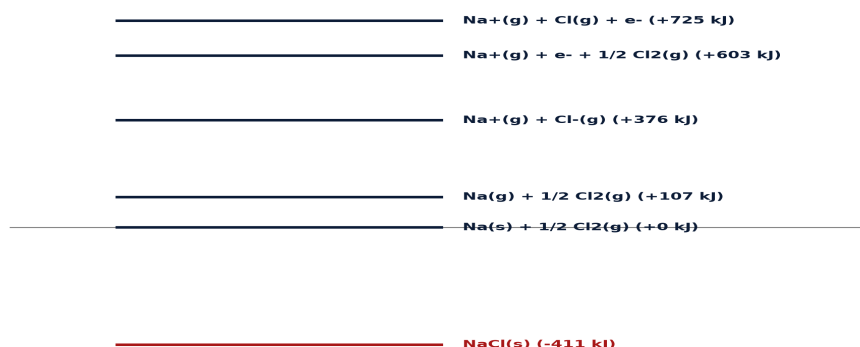
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M₂X (where M is divalent). [1]

Q8. 8. Born-Haber & Lattice Energy Step 8**[3 Marks]****WCH14/01/Sample/Q8**

Consider ionic compound 8 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

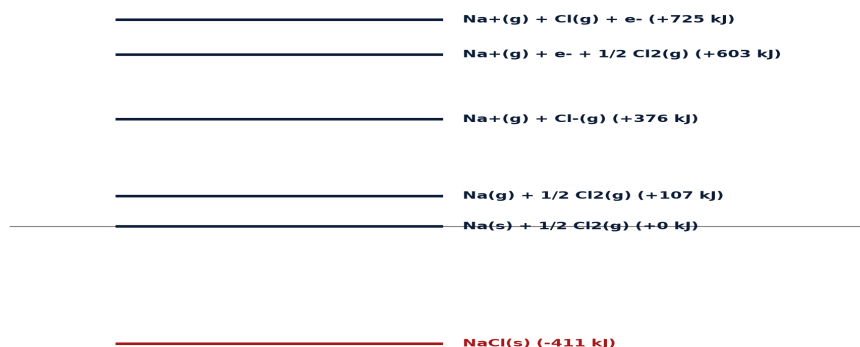
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M₂X (where M is divalent). [1]

Q9. 9. Born-Haber & Lattice Energy Step 9**[3 Marks]****WCH14/01/sample/Q9**

Consider ionic compound 9 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

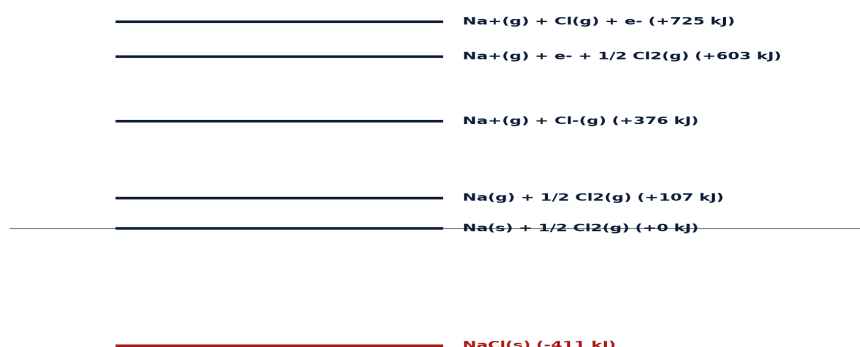
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M_2X (where M is divalent). [1]

Q10. 10. Born-Haber & Lattice Energy Step 10**[3 Marks]****WCH14/01/sample/Q10**

Consider ionic compound 10 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

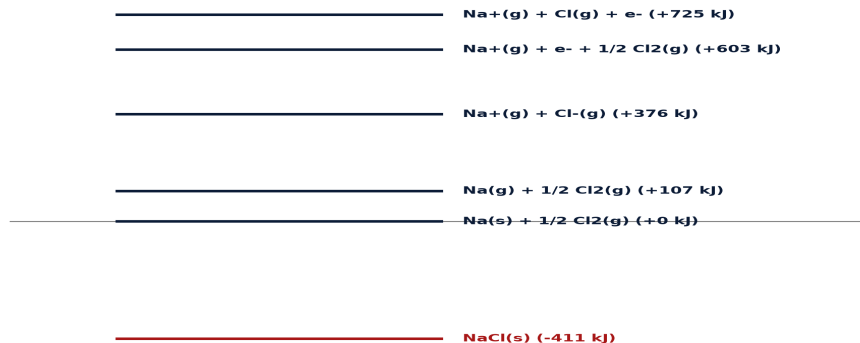
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M_2X (where M is divalent). [1]

Q11. 11. Born-Haber & Lattice Energy Step 11**[3 Marks]****WCH14/01/Sample/Q11**

Consider ionic compound 11 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

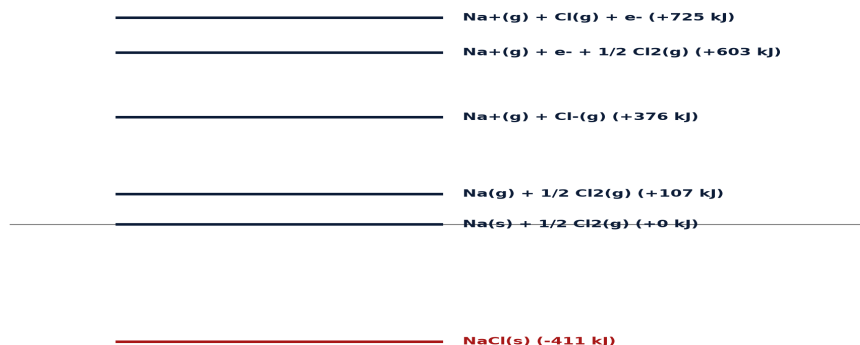
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M₂X (where M is divalent). [1]

Q12. 12. Born-Haber & Lattice Energy Step 12**[3 Marks]****WCH14/01/Sample/Q12**

Consider ionic compound 12 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

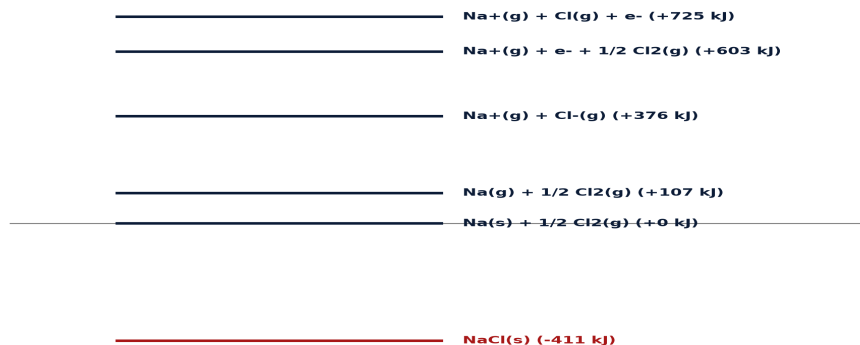
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M₂X (where M is divalent). [1]

Q13. 13. Born-Haber & Lattice Energy Step 13**[3 Marks]****WCH14/01/Sample/Q13**

Consider ionic compound 13 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

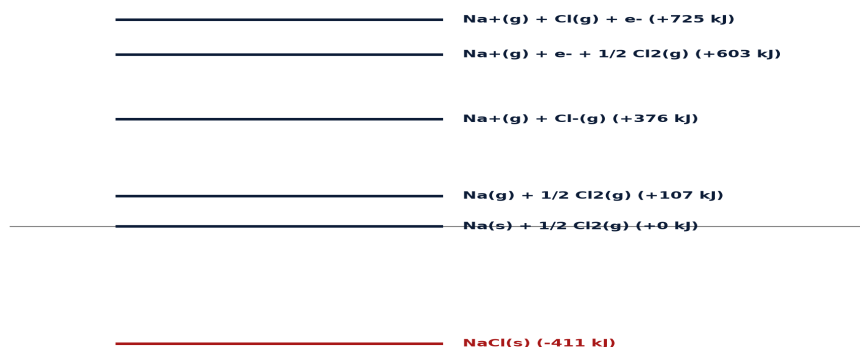
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M₂X (where M is divalent). [1]

Q14. 14. Born-Haber & Lattice Energy Step 14**[3 Marks]****WCH14/01/Sample/Q14**

Consider ionic compound 14 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

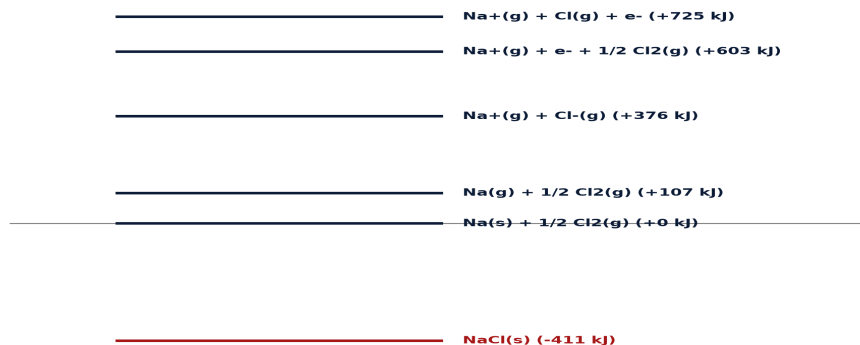
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M₂X (where M is divalent). [1]

Q15. 15. Born-Haber & Lattice Energy Step 15**[3 Marks]****WCH14/01/Sample/Q15**

Consider ionic compound 15 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

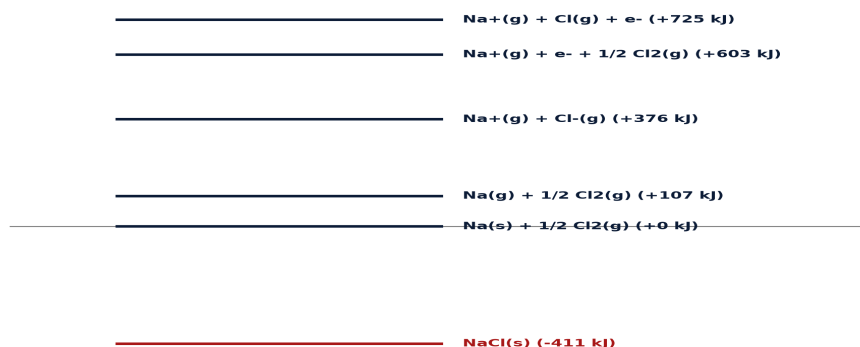
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M₂X (where M is divalent). [1]

Q16. 16. Born-Haber & Lattice Energy Step 16**[3 Marks]****WCH14/01/Sample/Q16**

Consider ionic compound 16 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

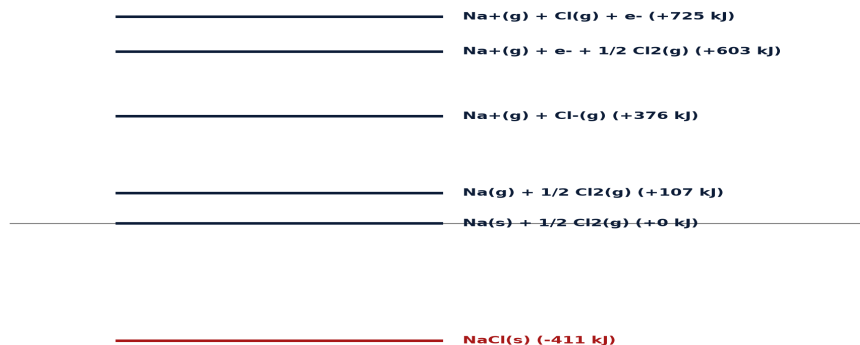
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M₂X (where M is divalent). [1]

Q17. 17. Born-Haber & Lattice Energy Step 17**[3 Marks]****WCH14/01/Sample/Q17**

Consider ionic compound 17 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

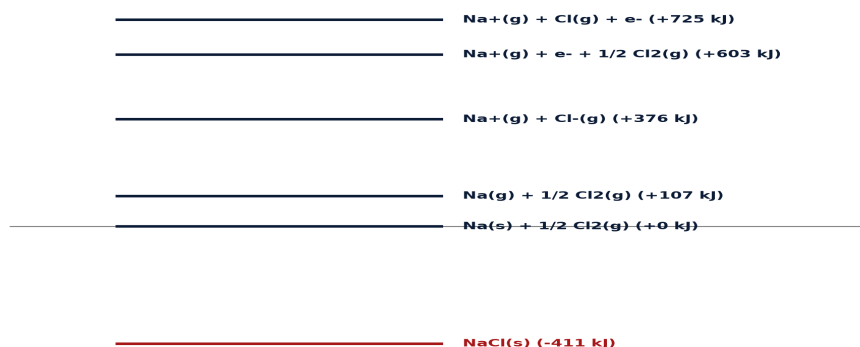
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M₂X (where M is divalent). [1]

Q18. 18. Born-Haber & Lattice Energy Step 18**[3 Marks]****WCH14/01/Sample/Q18**

Consider ionic compound 18 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

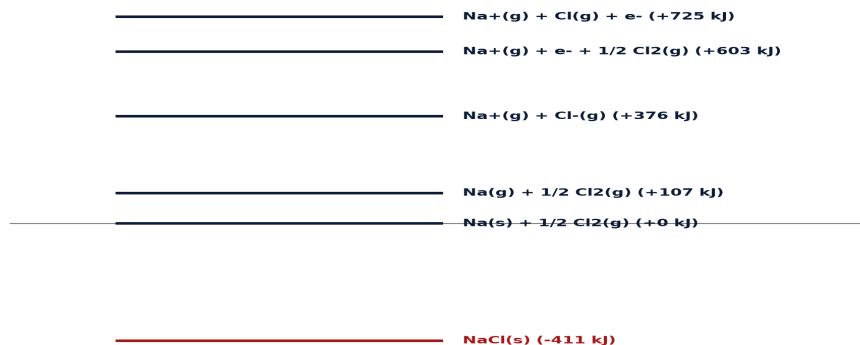
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M₂X (where M is divalent). [1]

Q19. 19. Born-Haber & Lattice Energy Step 19**[3 Marks]****WCH14/01/Sample/Q19**

Consider ionic compound 19 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

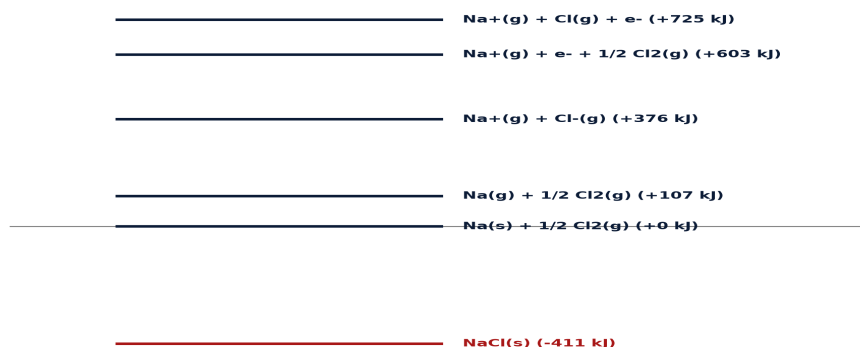
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M₂X (where M is divalent). [1]

Q20. 20. Born-Haber & Lattice Energy Step 20**[3 Marks]****WCH14/01/Sample/Q20**

Consider ionic compound 20 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

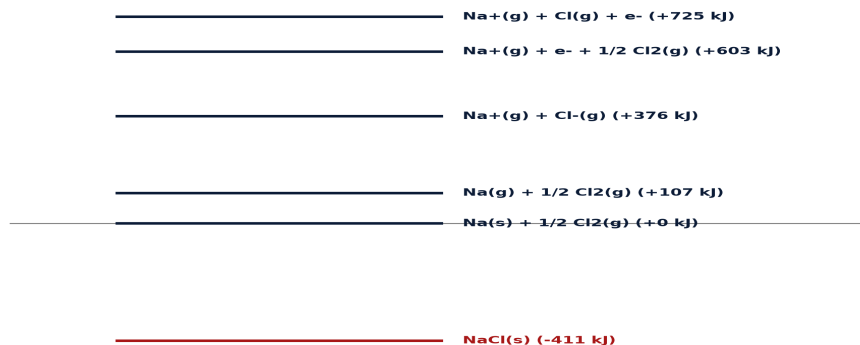
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M₂X (where M is divalent). [1]

Q21. 21. Born-Haber & Lattice Energy Step 21**[3 Marks]****WCH14/01/Sample/Q21**

Consider ionic compound 21 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

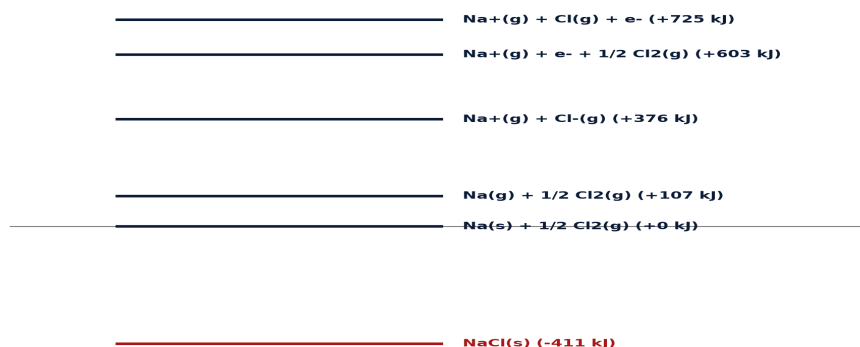
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M₂X (where M is divalent). [1]

Q22. 22. Born-Haber & Lattice Energy Step 22**[3 Marks]****WCH14/01/Sample/Q22**

Consider ionic compound 22 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

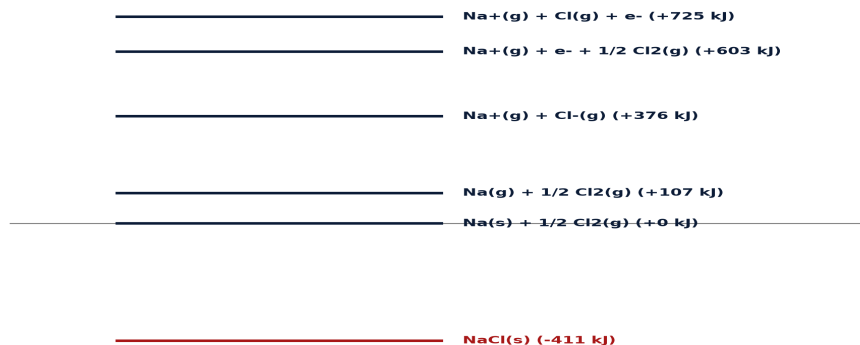
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M₂X (where M is divalent). [1]

Q23. 23. Born-Haber & Lattice Energy Step 23**[3 Marks]****WCH14/01/Sample/Q23**

Consider ionic compound 23 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

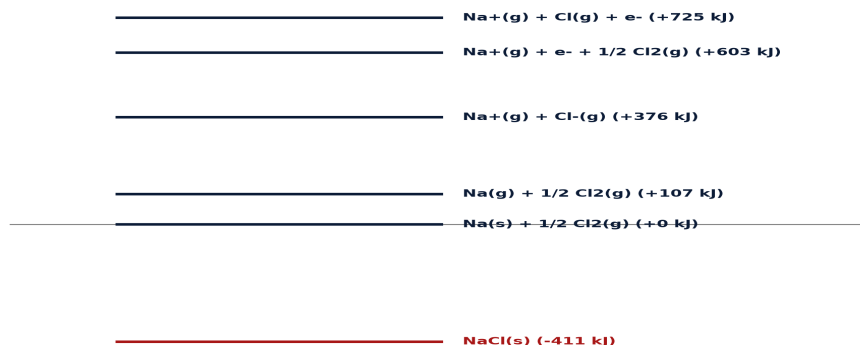
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M₂X (where M is divalent). [1]

Q24. 24. Born-Haber & Lattice Energy Step 24**[3 Marks]****WCH14/01/Sample/Q24**

Consider ionic compound 24 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

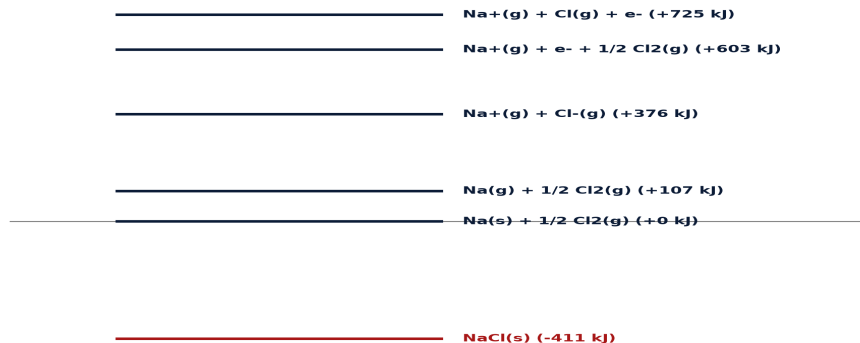
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M₂X (where M is divalent). [1]

Q25. 25. Born-Haber & Lattice Energy Step 25**[3 Marks]****WCH14/01/Sample/Q25**

Consider ionic compound 25 (MX): $\Delta_f H = -500 \text{ kJ mol}^{-1}$, sum of ionization and atomization enthalpies = $+800 \text{ kJ mol}^{-1}$, electron affinity = -350 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Calculate the lattice energy of MX. [2]

(b) Compare the lattice energy of MX with M₂X (where M is divalent). [1]

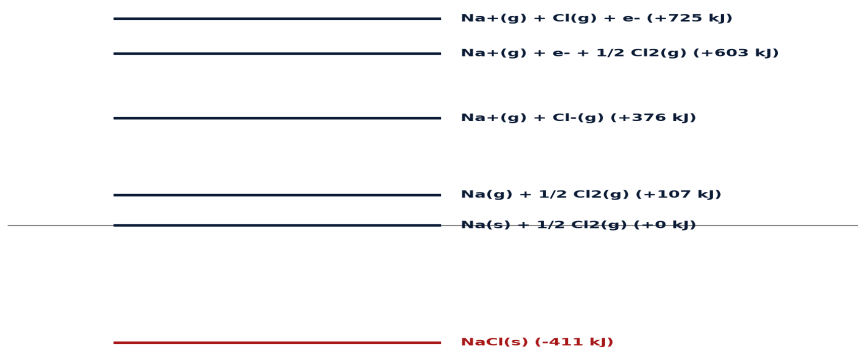
SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50

Q26. 26. A* Challenge: Experimental vs Theoretical Lattice & Polarization 26

[5 Marks]
WCH14/01/Hard/Q26

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)



(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

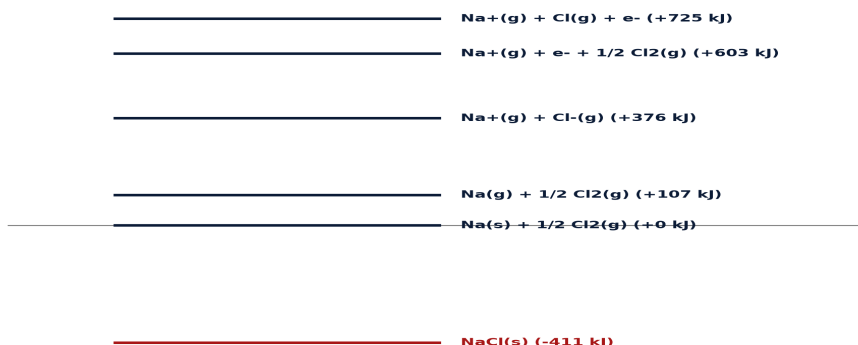
(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q27. 27. A* Challenge: Experimental vs Theoretical Lattice & Polarization 27

[5 Marks]
WCH14/01/Hard/Q27

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)



(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

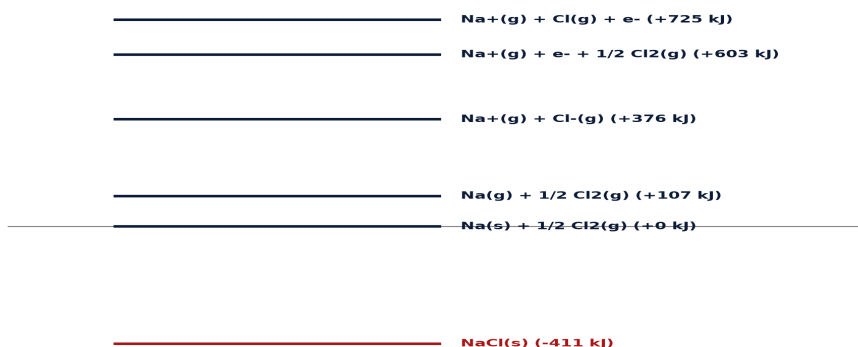
(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q28. 28. A* Challenge: Experimental vs Theoretical Lattice & Polarization 28

[5 Marks]
WCH14/01/Hard/Q28

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)



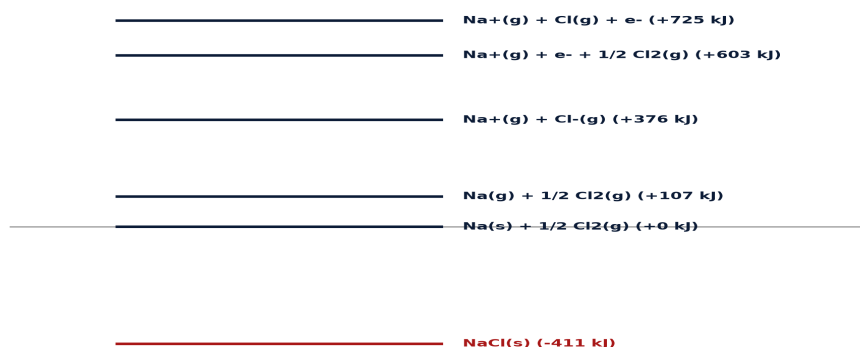
(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q29. 29. A* Challenge: Experimental vs Theoretical Lattice & Polarization 29

[5 Marks]
WCH14/01/Hard/Q29

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)

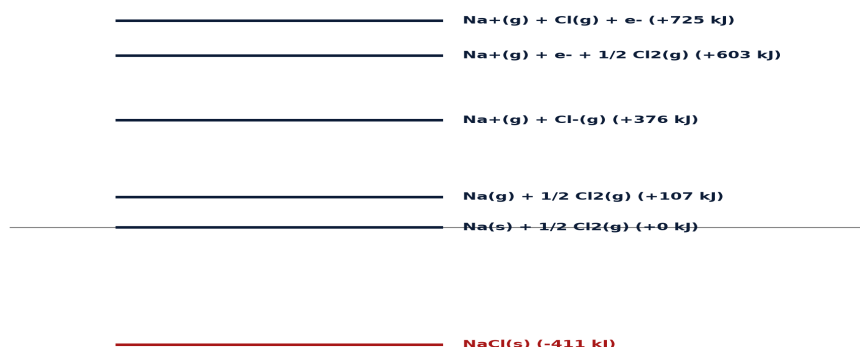
(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q30. 30. A* Challenge: Experimental vs Theoretical Lattice & Polarization 30

[5 Marks]
WCH14/01/Hard/Q30

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)

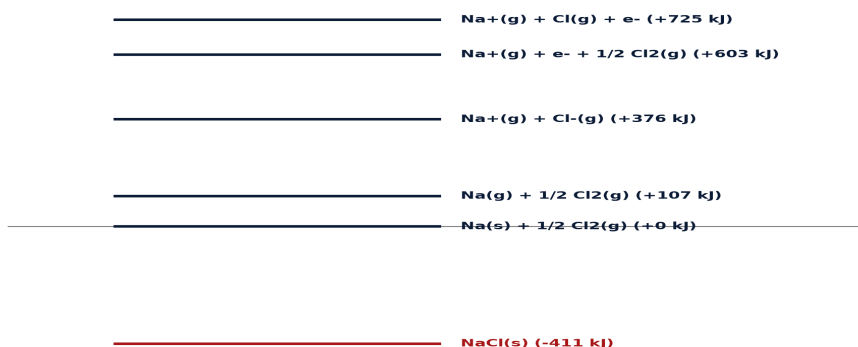
(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q31. 31. A* Challenge: Experimental vs Theoretical Lattice & Polarization 31

[5 Marks]
WCH14/01/Hard/Q31

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)


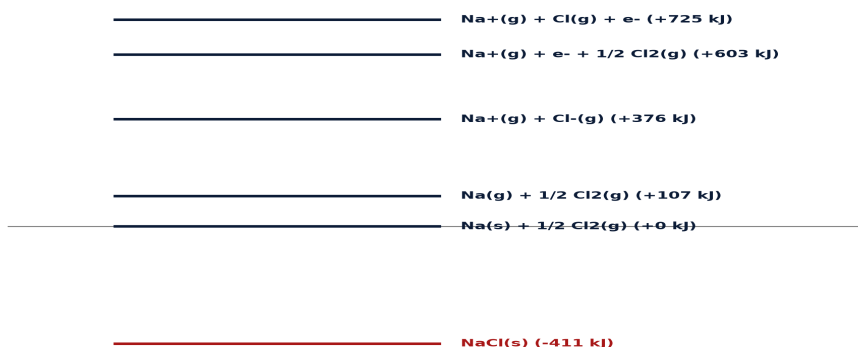
(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q32. 32. A* Challenge: Experimental vs Theoretical Lattice & Polarization 32

[5 Marks]
WCH14/01/Hard/Q32

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)


(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

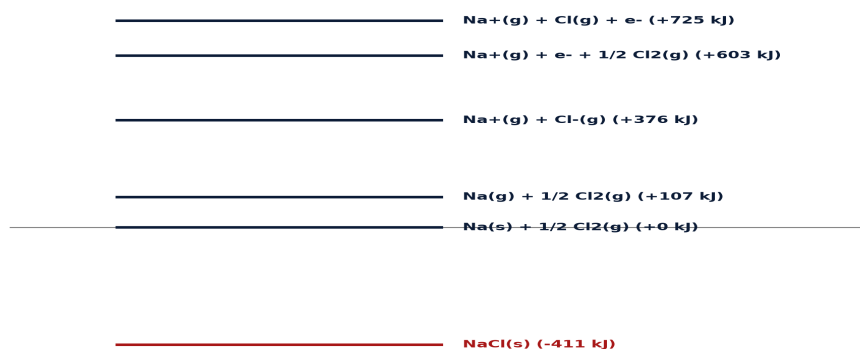
Q33. 33. A* Challenge: Experimental vs Theoretical Lattice & Polarization 33

[5 Marks]

WCH14/01/Hard/Q33

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)



(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

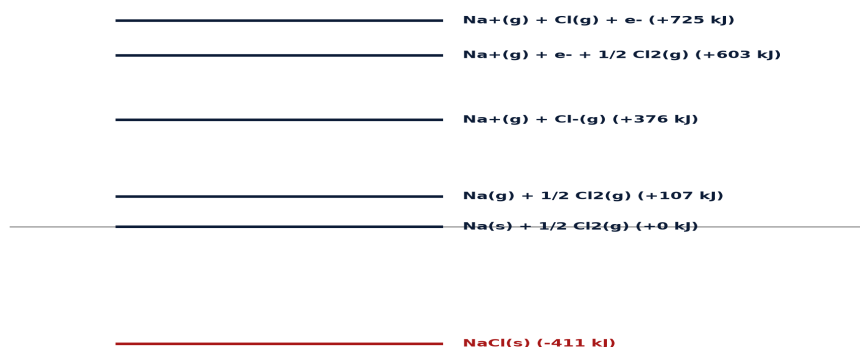
(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q34. 34. A* Challenge: Experimental vs Theoretical Lattice & Polarization 34

[5 Marks]

WCH14/01/Hard/Q34

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)

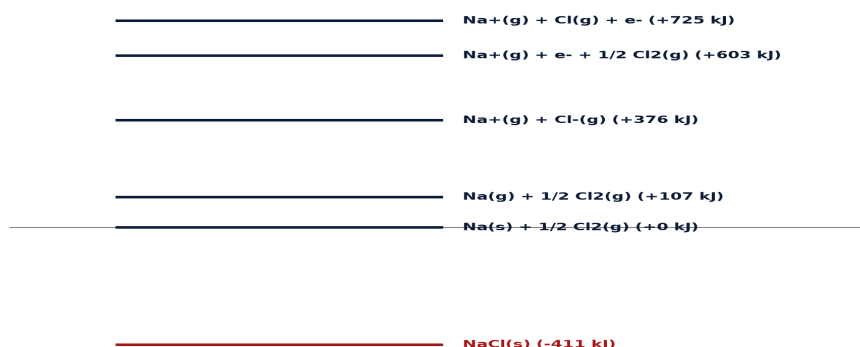
(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q35. 35. A* Challenge: Experimental vs Theoretical Lattice & Polarization 35

[5 Marks]
WCH14/01/Hard/Q35

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)

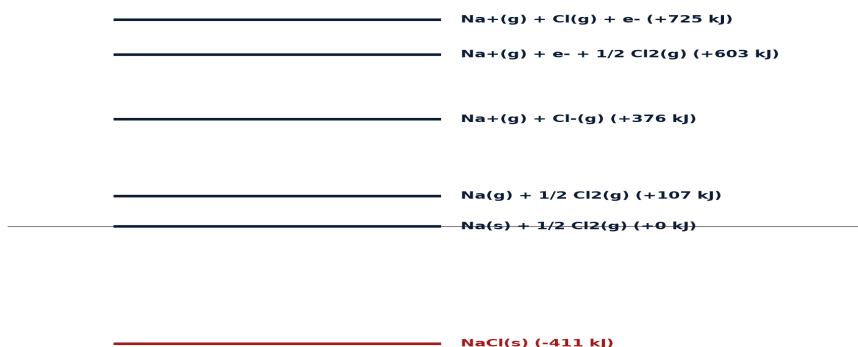
(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q36. 36. A* Challenge: Experimental vs Theoretical Lattice & Polarization 36

[5 Marks]
WCH14/01/Hard/Q36

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)


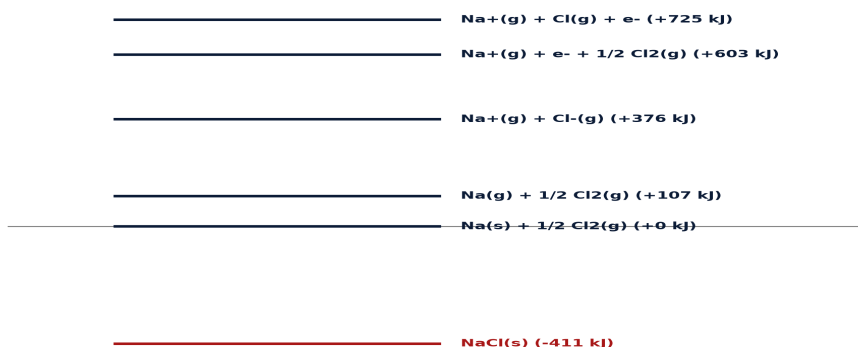
(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q37. 37. A* Challenge: Experimental vs Theoretical Lattice & Polarization 37

[5 Marks]
WCH14/01/Hard/Q37

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)


(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

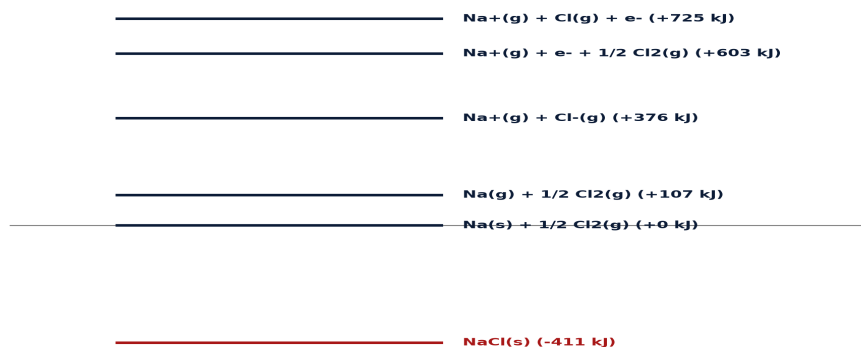
Q38. 38. A* Challenge: Experimental vs Theoretical Lattice & Polarization 38

[5 Marks]

WCH14/01/Hard/Q38

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)



(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

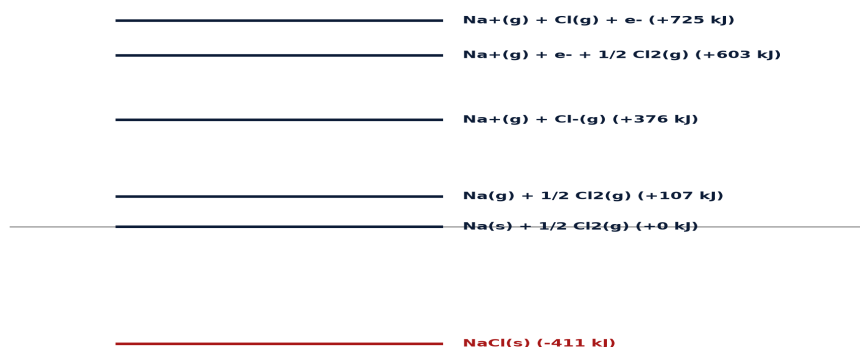
(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q39. 39. A* Challenge: Experimental vs Theoretical Lattice & Polarization 39

[5 Marks]

WCH14/01/Hard/Q39

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

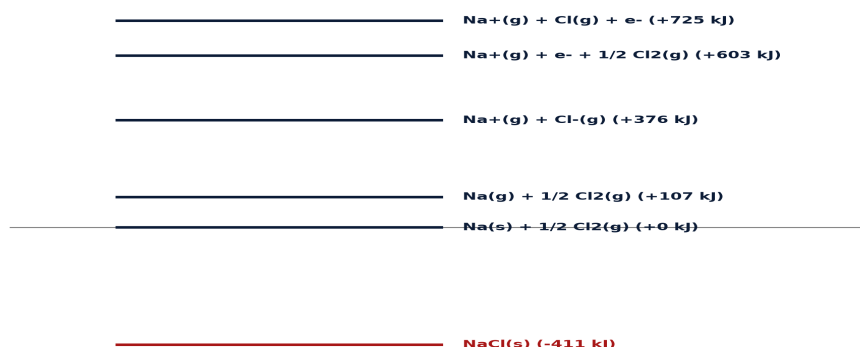
(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q40. 40. A* Challenge: Experimental vs Theoretical Lattice & Polarization 40

[5 Marks]

WCH14/01/Hard/Q40

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

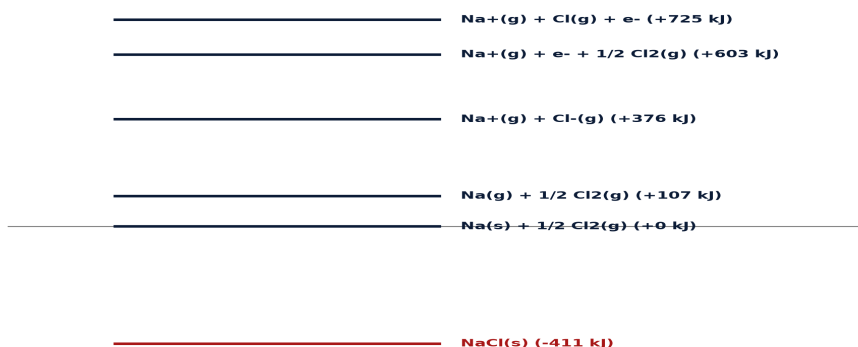
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q41. 41. A* Challenge: Experimental vs Theoretical Lattice & Polarization 41**[5 Marks]**
WCH14/01/Hard/Q41

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

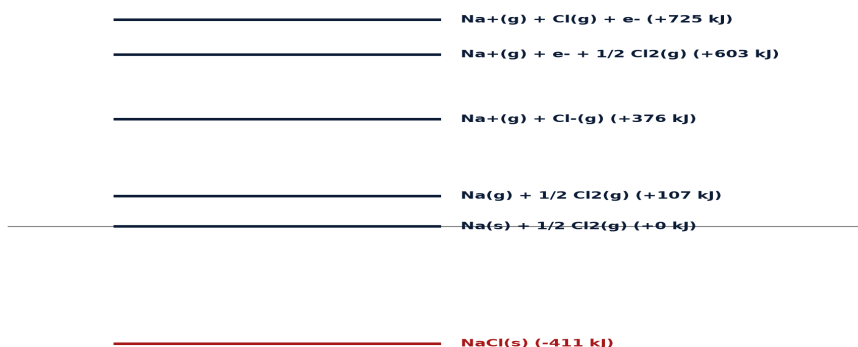
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q42. 42. A* Challenge: Experimental vs Theoretical Lattice & Polarization 42**[5 Marks]**
WCH14/01/Hard/Q42

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

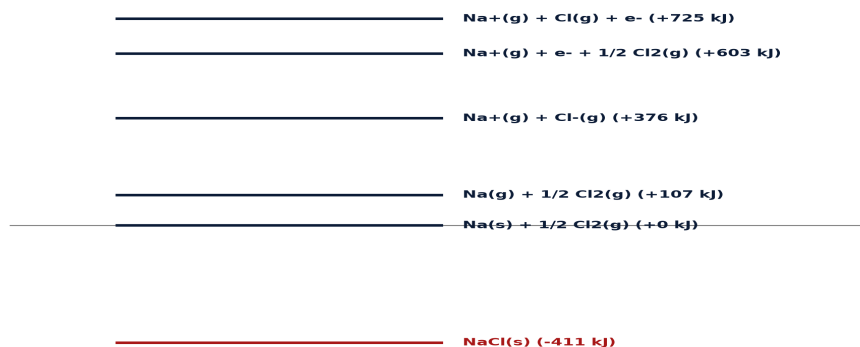
Q43. 43. A* Challenge: Experimental vs Theoretical Lattice & Polarization 43

[5 Marks]

WCH14/01/Hard/Q43

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)



(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

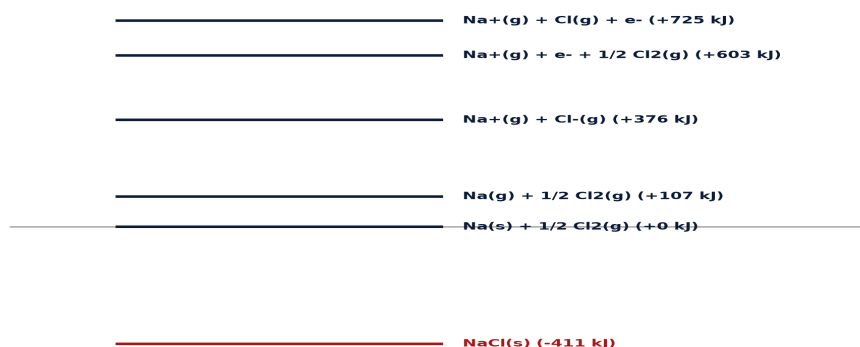
(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q44. 44. A* Challenge: Experimental vs Theoretical Lattice & Polarization 44

[5 Marks]

WCH14/01/Hard/Q44

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)

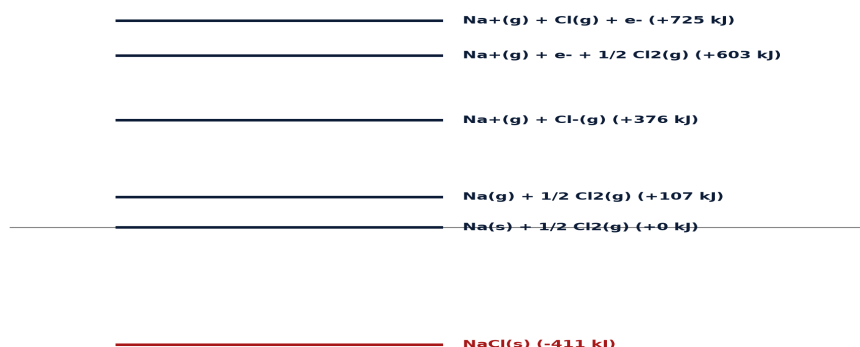
(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q45. 45. A* Challenge: Experimental vs Theoretical Lattice & Polarization 45

[5 Marks]
WCH14/01/Hard/Q45

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

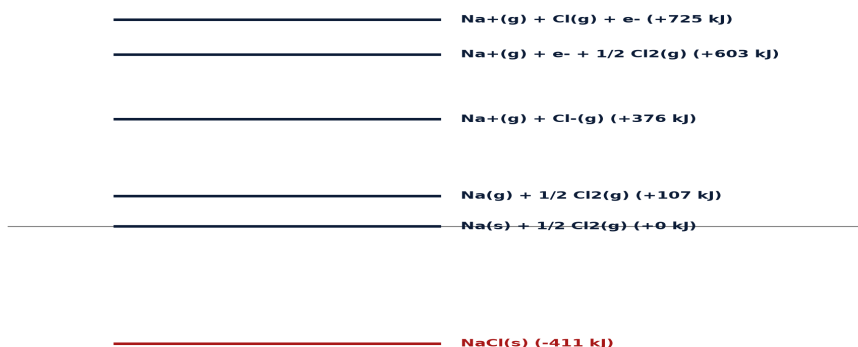
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q46. 46. A* Challenge: Experimental vs Theoretical Lattice & Polarization 46**[5 Marks]**
WCH14/01/Hard/Q46

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

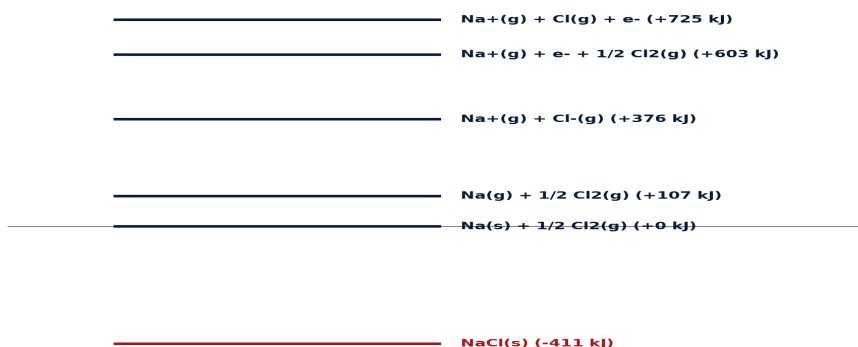
Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q47. 47. A* Challenge: Experimental vs Theoretical Lattice & Polarization 47**[5 Marks]**
WCH14/01/Hard/Q47

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

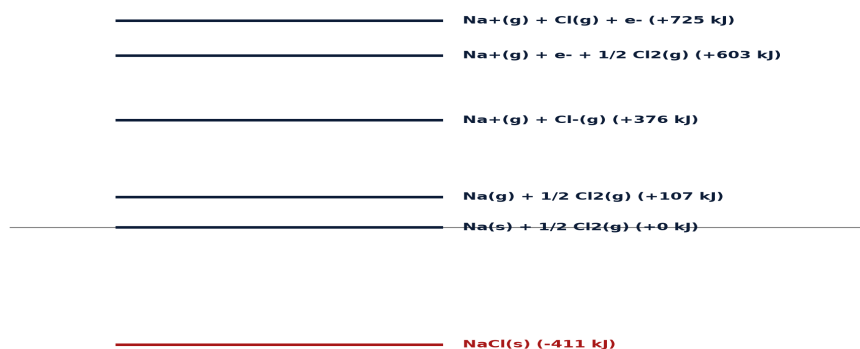
Q48. 48. A* Challenge: Experimental vs Theoretical Lattice & Polarization 48

[5 Marks]

WCH14/01/Hard/Q48

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)



(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

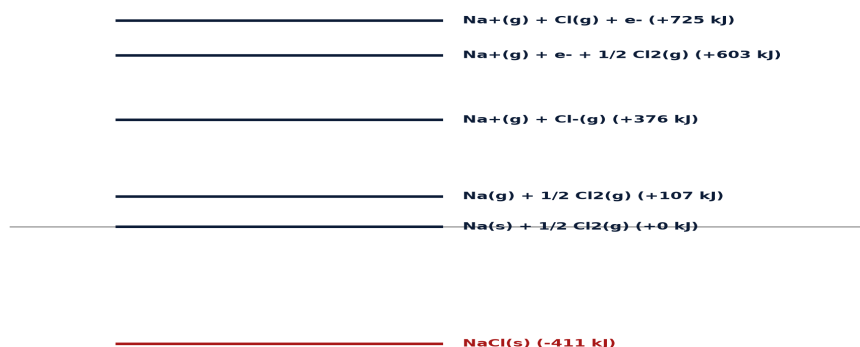
(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q49. 49. A* Challenge: Experimental vs Theoretical Lattice & Polarization 49

[5 Marks]

WCH14/01/Hard/Q49

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

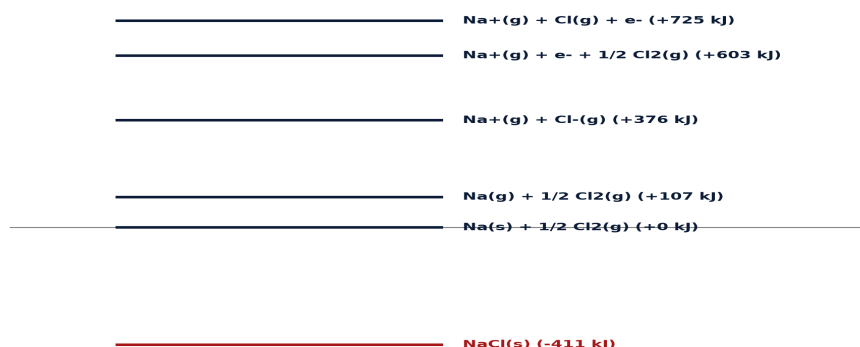
(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

Q50. 50. A* Challenge: Experimental vs Theoretical Lattice & Polarization 50

[5 Marks]

WCH14/01/Hard/Q50

For silver iodide, AgI: Experimental lattice energy = -887 kJ mol^{-1} . Theoretical Born-Landé lattice energy = -770 kJ mol^{-1} .

Born-Haber Cycle for Sodium Chloride (NaCl)

(a) Explain the difference between experimental and theoretical lattice energy for AgI. [3]

(b) Apply Hess's law to calculate the enthalpy of solution of AgI given $\Delta_{\text{hyd}} H(\text{Ag}^+) = -464$ and $\Delta_{\text{hyd}} H(\text{I}^-) = -293 \text{ kJ mol}^{-1}$. [2]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 12B

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Lattice Energy Definition 1. (a) Enthalpy change when one mole of a solid ionic lattice is formed from its constituent gaseous ions (1) under standard conditions (1). 1. (b) Exothermic (negative) (1).	[3]
Q2	Q2: 2. Born-Haber Cycle Calculation for NaCl 2. (a) Correct cycle diagram showing $\text{Na(s)} + \frac{1}{2}\text{Cl}_2\text{(g)} \rightarrow \text{Na(g)} + \text{Cl(g)} \rightarrow \text{NaCl(s)}$ (2). 2. (b) $\Delta_f H = \Delta_{\text{atom}} H(\text{Na}) + \text{IE}(\text{Na}) + \Delta_{\text{atom}} H(\text{Cl}) + \text{EA}(\text{Cl}) + \Delta_{\text{LE}} H$ (1). $-411 = +107 + 496 + 122 - 349 + \Delta_{\text{LE}} H$ (1). $\Delta_{\text{LE}} H = -411 - 376 = -787 \text{ kJ mol}^{-1}$ (1).	[5]
Q3	Q3: 3. Born-Haber & Lattice Energy Step 3 3. (a) $\Delta_{\text{LE}} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 3. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q4	Q4: 4. Born-Haber & Lattice Energy Step 4 4. (a) $\Delta_{\text{LE}} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 4. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q5	Q5: 5. Born-Haber & Lattice Energy Step 5 5. (a) $\Delta_{\text{LE}} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 5. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q6	Q6: 6. Born-Haber & Lattice Energy Step 6 6. (a) $\Delta_{\text{LE}} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 6. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q7	Q7: 7. Born-Haber & Lattice Energy Step 7 7. (a) $\Delta_{\text{LE}} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 7. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q8	Q8: 8. Born-Haber & Lattice Energy Step 8 8. (a) $\Delta_{\text{LE}} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 8. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q9	Q9: 9. Born-Haber & Lattice Energy Step 9 9. (a) $\Delta_{\text{LE}} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 9. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q10	Q10: 10. Born-Haber & Lattice Energy Step 10 10. (a) $\Delta_{\text{LE}} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 10. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q11	Q11: 11. Born-Haber & Lattice Energy Step 11 11. (a) $\Delta_{\text{LE}} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 11. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q12	Q12: 12. Born-Haber & Lattice Energy Step 12 12. (a) $\Delta_{\text{LE}} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 12. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]

Q13	Q13: 13. Born-Haber & Lattice Energy Step 13 13. (a) $\Delta_{LE} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 13. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q14	Q14: 14. Born-Haber & Lattice Energy Step 14 14. (a) $\Delta_{LE} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 14. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q15	Q15: 15. Born-Haber & Lattice Energy Step 15 15. (a) $\Delta_{LE} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 15. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q16	Q16: 16. Born-Haber & Lattice Energy Step 16 16. (a) $\Delta_{LE} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 16. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q17	Q17: 17. Born-Haber & Lattice Energy Step 17 17. (a) $\Delta_{LE} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 17. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q18	Q18: 18. Born-Haber & Lattice Energy Step 18 18. (a) $\Delta_{LE} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 18. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q19	Q19: 19. Born-Haber & Lattice Energy Step 19 19. (a) $\Delta_{LE} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 19. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q20	Q20: 20. Born-Haber & Lattice Energy Step 20 20. (a) $\Delta_{LE} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 20. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q21	Q21: 21. Born-Haber & Lattice Energy Step 21 21. (a) $\Delta_{LE} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 21. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q22	Q22: 22. Born-Haber & Lattice Energy Step 22 22. (a) $\Delta_{LE} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 22. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q23	Q23: 23. Born-Haber & Lattice Energy Step 23 23. (a) $\Delta_{LE} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 23. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q24	Q24: 24. Born-Haber & Lattice Energy Step 24 24. (a) $\Delta_{LE} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 24. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]
Q25	Q25: 25. Born-Haber & Lattice Energy Step 25 25. (a) $\Delta_{LE} H = -500 - (800 - 350) = -950 \text{ kJ mol}^{-1}$ (2). 25. (b) Divalent cation has higher charge density $\rightarrow$ stronger electrostatic attraction $\rightarrow$ much more exothermic lattice energy (1).	[3]

Q26	Q26: 26. A* Challenge: Experimental vs Theoretical Lattice & Polarization 26 26. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 26. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q27	Q27: 27. A* Challenge: Experimental vs Theoretical Lattice & Polarization 27 27. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 27. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q28	Q28: 28. A* Challenge: Experimental vs Theoretical Lattice & Polarization 28 28. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 28. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q29	Q29: 29. A* Challenge: Experimental vs Theoretical Lattice & Polarization 29 29. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 29. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q30	Q30: 30. A* Challenge: Experimental vs Theoretical Lattice & Polarization 30 30. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 30. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q31	Q31: 31. A* Challenge: Experimental vs Theoretical Lattice & Polarization 31 31. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 31. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q32	Q32: 32. A* Challenge: Experimental vs Theoretical Lattice & Polarization 32 32. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 32. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q33	Q33: 33. A* Challenge: Experimental vs Theoretical Lattice & Polarization 33 33. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 33. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q34	Q34: 34. A* Challenge: Experimental vs Theoretical Lattice & Polarization 34 34. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 34. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]

Q35	Q35: 35. A* Challenge: Experimental vs Theoretical Lattice & Polarization 35 35. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 35. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q36	Q36: 36. A* Challenge: Experimental vs Theoretical Lattice & Polarization 36 36. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 36. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q37	Q37: 37. A* Challenge: Experimental vs Theoretical Lattice & Polarization 37 37. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 37. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q38	Q38: 38. A* Challenge: Experimental vs Theoretical Lattice & Polarization 38 38. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 38. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q39	Q39: 39. A* Challenge: Experimental vs Theoretical Lattice & Polarization 39 39. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 39. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q40	Q40: 40. A* Challenge: Experimental vs Theoretical Lattice & Polarization 40 40. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 40. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q41	Q41: 41. A* Challenge: Experimental vs Theoretical Lattice & Polarization 41 41. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 41. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q42	Q42: 42. A* Challenge: Experimental vs Theoretical Lattice & Polarization 42 42. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 42. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q43	Q43: 43. A* Challenge: Experimental vs Theoretical Lattice & Polarization 43 43. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 43. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]

Q44	Q44: 44. A* Challenge: Experimental vs Theoretical Lattice & Polarization 44 44. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 44. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q45	Q45: 45. A* Challenge: Experimental vs Theoretical Lattice & Polarization 45 45. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 45. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q46	Q46: 46. A* Challenge: Experimental vs Theoretical Lattice & Polarization 46 46. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 46. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q47	Q47: 47. A* Challenge: Experimental vs Theoretical Lattice & Polarization 47 47. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 47. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q48	Q48: 48. A* Challenge: Experimental vs Theoretical Lattice & Polarization 48 48. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 48. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q49	Q49: 49. A* Challenge: Experimental vs Theoretical Lattice & Polarization 49 49. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 49. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]
Q50	Q50: 50. A* Challenge: Experimental vs Theoretical Lattice & Polarization 50 50. (a) Theoretical model assumes purely ionic spherical ions (1). Experimental > theoretical indicates significant covalent character (1) due to polarization of large iodide anion by small polarising Ag ⁺ cation (1). 50. (b) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Ag}^+) + \Delta_{\text{hyd}} H(\text{I}^-) - \Delta_{\text{LE}} H = -464 - 293 - (-887) = +130 \text{ kJ mol}^{-1}$ (2).	[5]

SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 12B

FAQ #1: Definition of Atomisation Enthalpy

Category: Born-Haber
Definitions

COMMON EXAMINER TRAP / MISTAKE:

Using 1 mole of element instead of forming 1 mole of gaseous atoms.

MODEL ANSWER / EXAMINER EXPECTATION:

Enthalpy of atomisation $\Delta_{\text{atom}} H$ is the enthalpy change when ONE MOLE OF GASEOUS ATOMS is formed from the element in its standard state (e.g. $\frac{1}{2} \text{Cl}_2(\text{g}) \rightarrow \text{Cl}(\text{g})$).

FAQ #2: First vs Second Electron Affinity Signs

Category: Electron Affinity
Signs

COMMON EXAMINER TRAP / MISTAKE:

Assuming all electron affinities are exothermic.

MODEL ANSWER / EXAMINER EXPECTATION:

First electron affinity (EA1) is EXOTHERMIC (-). Second electron affinity (EA2) is ENDOTHERMIC (+) because an electron is added to a negatively charged anion (electrostatic repulsion).

FAQ #3: Born-Haber Cycle Direction Arrows

Category: Hess's Law Cycle

COMMON EXAMINER TRAP / MISTAKE:

Drawing cycle arrows in random directions.

MODEL ANSWER / EXAMINER EXPECTATION:

Standard enthalpies of formation go DOWN from elements $\rightarrow$ compound. Atomisation and IE go UP $\rightarrow$ ions. EA goes DOWN $\rightarrow$ anions. Lattice energy goes DOWN from gaseous ions $\rightarrow$ solid lattice.

FAQ #4: Experimental vs Theoretical Lattice Energy Difference

Category: Polarization /
Covalency

COMMON EXAMINER TRAP / MISTAKE:

Attributing difference in lattice energy to experimental measurement error.

MODEL ANSWER / EXAMINER EXPECTATION:

If experimental LE is significantly MORE EXOTHERMIC than theoretical (Born-Landé), the bonding has COVALENT CHARACTER caused by cation polarization of the anion (Fajans' rules).

FAQ #5: Polarising Power vs Polarisability

Category: Fajans' Rules

COMMON EXAMINER TRAP / MISTAKE:

Confusing polarising power of cations with polarisability of anions.

MODEL ANSWER / EXAMINER EXPECTATION:

Cations have POLARISING POWER (high charge, small ionic radius). Anions are POLARISABLE (large ionic radius, electron cloud easily distorted).

FAQ #6: Enthalpy of Solution Formula

Category: Hydration vs Lattice

COMMON EXAMINER TRAP / MISTAKE:

Subtracting hydration enthalpies from lattice energy in reverse order.

MODEL ANSWER / EXAMINER EXPECTATION:

$\Delta_{\text{sol}} H = \text{Sum } \Delta_{\text{hyd}} H(\text{ions}) - \Delta_{\text{LE}} H$ (where LE is defined as lattice formation).
Alternatively: $\Delta_{\text{sol}} H = \Delta_{\text{dissociation}} H + \text{Sum } \Delta_{\text{hyd}} H$.

FAQ #7: Hydration Enthalpy Factors**Category: Ion-Dipole Attraction****COMMON EXAMINER TRAP / MISTAKE:**

Stating larger ions have more exothermic hydration enthalpy.

MODEL ANSWER / EXAMINER EXPECTATION:

Hydration enthalpy depends on CHARGE DENSITY: smaller ionic radius and higher charge → higher charge density → stronger ion-dipole attraction to water → MORE EXOTHERMIC $\Delta_{\text{hyd}} H$.

FAQ #8: Group 2 Sulfate Solubility Trend**Category: Solubility Trends****COMMON EXAMINER TRAP / MISTAKE:**

Explaining group 2 sulfate solubility using lattice energy alone.

MODEL ANSWER / EXAMINER EXPECTATION:

Down Group 2, cation radius increases. Lattice energy decreases slightly, but hydration enthalpy of cation decreases much more rapidly. Net $\Delta_{\text{sol}} H$ becomes more positive → solubility DECREASES.

FAQ #9: Group 2 Hydroxide Solubility Trend**Category: Solubility Trends****COMMON EXAMINER TRAP / MISTAKE:**

Applying sulfate solubility trend to hydroxides.

MODEL ANSWER / EXAMINER EXPECTATION:

OH^- is a small anion. Down Group 2, lattice energy decreases faster than hydration enthalpy. Net $\Delta_{\text{sol}} H$ becomes less positive / more negative → hydroxide solubility INCREASES down the group.

FAQ #10: Lattice Dissociation vs Lattice Formation**Category: Enthalpy Definitions****COMMON EXAMINER TRAP / MISTAKE:**

Confusing lattice formation ($-\text{LE}$) with lattice dissociation ($+\text{LE}$).

MODEL ANSWER / EXAMINER EXPECTATION:

Lattice FORMATION enthalpy: gaseous ions → solid lattice (EXOTHERMIC, negative). Lattice DISSOCIATION enthalpy: solid lattice → gaseous ions (ENDOTHERMIC, positive). Always check the sign in data tables.